

Remote sensing of vegetation responses to flooding of a semi-arid floodplain: Implications for monitoring ecological effects of environmental flows

Neil C. Sims^{a,*}, Matthew J. Colloff^b

^a CSIRO Land and Water, Private Bag 10, Clayton, Victoria 3169, Australia

^b CSIRO Ecosystem Sciences, GPO Box 1700, Canberra, ACT 2601, Australia

ARTICLE INFO

Article history:

Received 7 December 2010

Received in revised form 28 October 2011

Accepted 11 December 2011

Keywords:

NDVI

MODIS

Primary productivity

Flood response

Paroo

Semi-arid

Australia

Murray-Darling Basin

Baseline

ABSTRACT

Environmental flows are used to support the structure and function of floodplain, wetland and riverine ecosystems that are subject to stress from drought, climate change or water resource development. The financial costs of environmental flows are considerable and it is therefore highly desirable that the ecological benefits are monitored. Productivity of floodplain vegetation is a critical ecological response to flooding, and represents an indicator of ecosystem 'health', being linked to structural integrity, habitat provision, nutrient cycling and numerous other ecosystem functions. Productivity of floodplain plant communities is hard to measure, being highly variable over time depending on seasonal and climatic factors and the frequency, magnitude and duration of flood events. We compared an index of plant growth – the Normalised Difference Vegetation Index (NDVI) calculated from MODIS satellite imagery – in the Paroo River Wetlands ecosystem, Australia, to NDVI in a nearby terrestrial area to quantify the magnitude and duration of floodplain vegetation responses to flooding alone. This analysis shows that a flood which inundated more than 50% of the Paroo River Wetlands in 2008 increased NDVI by up to 19% above non-flood levels and for a period of 13 months following flood recession. This approach has applications in planning environmental flows and in monitoring the ecological responses.

Crown Copyright © 2011 Published by Elsevier Ltd. All rights reserved.

1. Introduction

Environmental flows – volumes of water released within regulated river systems for ecological benefits – are used in many countries to maintain and enhance the ecological character and functions of floodplain, wetland and riverine ecosystems that may be subject to stress from drought, climate change or water resource development (IUCN, 2011). The real financial cost and the opportunity cost of acquiring and using water for environmental purposes is very high, and there is a clear need to measure the ecological benefits. Event-based monitoring of ecological responses to environmental flows is costly and time-consuming and there is a need to develop ecologically meaningful indicators that can be assessed consistently across floodplains and wetlands in a cost-effective, routine and reproducible manner.

One of the most evident and important ecological responses to flooding is an increase in primary productivity. Floodplain and wetland vegetation represents important habitat for fauna (Briggs and Thornton, 1999; Bodie and Semlitsch, 2000; Sheldon et al., 2002) and provides carbon and nutrient inputs that support terrestrial and aquatic food webs (Thorpe and Delong, 1994; Bunn et al., 2003).

Floodplains and wetlands tend to have higher levels of primary productivity than their terrestrial counterparts due to increased water availability and the disturbance effects of flooding (Junk et al., 1989; Junk and Piedade, 1993; Ogden et al., 2002), but their productivity levels are also typically highly variable over time due to the influence of these additional factors.

Floodplain primary productivity can be interpreted from satellite image data using the Normalised Difference Vegetation Index (NDVI). Pixel values in NDVI images have been correlated with many physical and physiological aspects of vegetation including primary productivity (Fuller, 1998), biomass (Hansen and Schjoerring, 2003), canopy structure and Leaf Area Index (Gamon et al., 1995) as well as species-richness of vascular plants and mammals (Oindo and Skidmore, 2002). The Moderate Resolution Imaging Spectroradiometer (MODIS) provides near-daily imagery of most parts of the surface of the world, including NDVI. Interpreting the contribution of flooding to a given NDVI value at a certain point in time is complicated, however, by the manifold influences on plant growth including rainfall, temperature and seasonality in addition to flooding.

The aim of this work is to identify the period of increased floodplain primary productivity attributable solely to flooding using MODIS NDVI images. This paper presents a method to minimise the influence of factors other than flooding on the interpretation of NDVI time-series data for floodplain and wetland ecosystems.

* Corresponding author: Tel.: +61 3 9545 2163; fax: +61 3 9545 2139.
E-mail address: Neil.Sims@csiro.au (N.C. Sims).

By comparing plant growth responses of a floodplain ecosystem to plant growth in a nearby terrestrial area that is not affected by flooding, plant growth responses due to rainfall or seasonal effects can be calibrated out from plant growth responses due to flooding.

2. Methods

2.1. Site characteristics

The study site was the Ramsar-listed Paroo River Wetlands in the semi-arid far north-western region of the Murray-Darling Basin (Fig. 1). This region contains some of the most important wetlands for waterbirds in Australia (Kingsford and Lee, 2010). Flooding of the Paroo River Wetlands occurs at least once a year on average, though its frequency is highly variable. Floods are generated from rainfall several hundred kilometres north in the upper catchment and via inflows to the southern part of the wetlands from the Warrego River to the east via the Cuttaburra Creek. Local heavy rainfalls may also inundate small wetlands and claypans.

The Paroo River is unregulated and represents a relatively natural hydrological system (CSIRO, 2007). It was chosen as a test site for this method, however, because most of the rest of the Murray-Darling Basin was in drought between 2000 and 2009 whereas flows in the Paroo were relatively unaffected (Colloff and Jin, 2009; Sims et al., 2011), and because the semi-arid nature of the Paroo means that post-flood NDVI strongly contrasts with pre-flood conditions. Spatial layers included in the Directory of Important Wetlands in Australia (DEWHA, 2005) were used to identify the boundary of Paroo River Wetlands.

2.2. Imagery used to estimate flooding and primary productivity

This analysis uses time series data calculated from MODIS images (MOD09A1 collection 5) captured between 18 February 2000 and 28 July 2009 showing surface reflectance at 500 m pixel resolution composited over an 8-day period. Only pixels that were wholly contained within the boundary of the wetland and baseline regions were included in this analysis. Data quality bands were not specifically interrogated, but compositing images over 8 days tends to reduce interference from cloud cover and sensor aberrations, resulting in increased image quality (USGS, 2010). Two products were developed from the MODIS data: an NDVI dataset of plant productivity and biomass, and an Open Water Likelihood (OWL) index representing the probability of standing water within a MODIS pixel (Kirby et al., 2008). NDVI values range from a theoretical minimum of -1 to a maximum of $+1$, with higher plant growth vigour and biomass being generally correlated with higher NDVI values. Further details of the calculation of OWL are given by Guerschman et al. (2009).

NDVI and OWL values were calculated from the Paroo River Wetlands and a terrestrial area approximately 5 km from the wetlands which does not experience flooding (Fig. 1). The purpose of data collected from the terrestrial area is to provide a baseline of NDVI change over time for local terrestrial vegetation. Variations in the baseline NDVI are assumed to be due to rainfall, changes in evaporation and evapotranspiration rates, seasonal and phenological changes, and variations in temperature. We assumed that floodplain vegetation is subject to these same influences, but with the additional effect of flooding. The ratio of change in NDVI in the baseline and floodplain regions should therefore be close to zero in the absence of flooding.

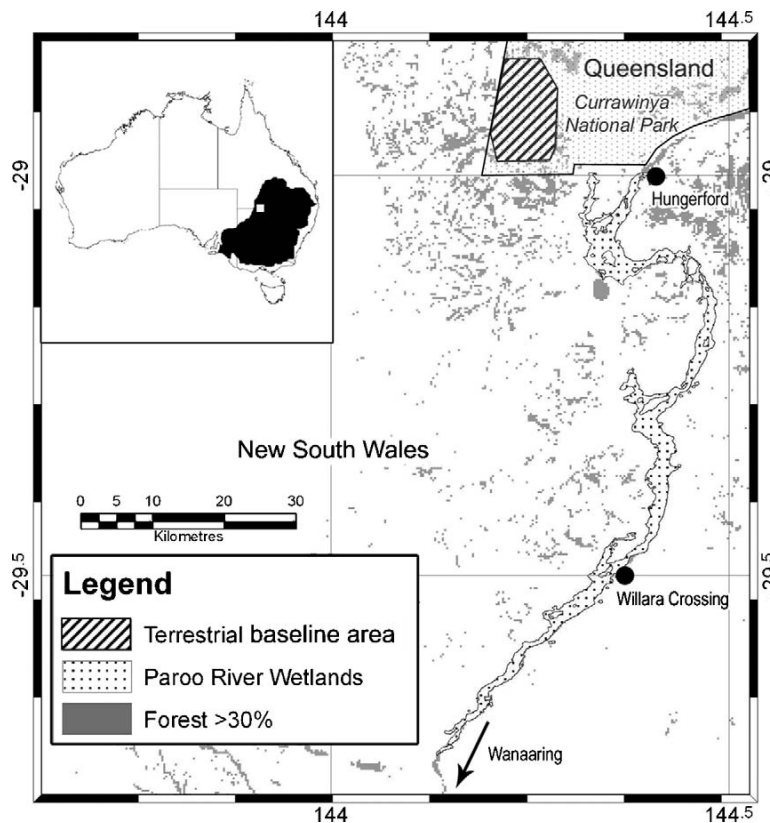


Fig. 1. The Paroo River Wetlands and the terrestrial baseline area, along with the boundary of Currawinya National Park. The inset shows the location of the Paroo River (white square) in the northern region of the Murray-Darling Basin (shown in black).

2.3. Vegetation in the wetland and baseline regions

Vegetation communities on floodplains are adapted to periodic flooding, and therefore tend to have a different plant species composition from terrestrial areas. Time series of NDVI are more influenced by plant growth form than by species, however (Donohue et al., 2009). Shallow rooted, short-lived non-forest vegetation tends to respond to flooding more rapidly but for a shorter duration than deep-rooted and longer lived tree types (Chesterfield, 1986). Each vegetation type makes a unique contribution to the NDVI pattern observed at the whole-ecosystem scale over the longer term. Locating the baseline region in a relatively natural area such as a National Park reduces the likelihood of land clearing or other disturbances which would introduce errors into the interpretation of vegetation response to flooding on the floodplain.

The terrestrial baseline in this study was located within the Currawinya National Park (Fig. 1). This area contains a similar proportion of 'forest' (actually, woodland, cf. below) and 'non-forest' land cover types as indicated in the National Carbon Accounting System Forest Extent and Change (v4) spatial data layer (DCC, 2008) and is subject to low levels of anthropogenic change as indicated by the Land Use of Australia – 2001/2002 spatial data layer (BRS, 2006). Assessment of the baseline and wetland areas using aerial and ground photography found both to consist of low open woodland. The wetlands contain black box (*Eucalyptus largifolens*) lignum (*Muehlenbeckia* spp.) and spike rush (*Eleocharis* spp.) swamps (Queensland Parks and Wildlife Service, 1999; Kingsford and Lee, 2010). Low open woodland in the National Park is dominated by mulga (*Acacia aneura*) and poplar box (*Eucalyptus populinea*) whereas in the wetland, the woodland component consists of gidgee (*Acacia cambagei*) and yapunyah (*Eucalyptus ochrophloia*) as the dominant species (Queensland Parks and Wildlife Service, 1999).

2.4. Calculation of the NDVI response due to flooding

The NDVI time series were mean-centred by dividing NDVI at each time step by the mean NDVI for all records of each time series. This accounts for differences in absolute NDVI between the baseline and wetlands. The wetland NDVI response due to flooding was then calculated from the mean-centred data as wetland NDVI minus baseline NDVI. The resulting time series is referred to as the 'NDVI anomaly' hereafter. NDVI anomaly values around zero indicate that the direction and magnitude of NDVI change in the baseline and wetland regions are similar. Positive values indicate that plant productivity in the wetlands is relatively higher than in the baseline zone compared to mean (normal) conditions, and vice versa.

3. Results

Fig. 2A shows the NDVI time series for the Paroo River Wetlands and the baseline area along with the proportion of the Paroo River Wetlands flooded. NDVI is generally higher in the wetlands (mean = 0.261; SD = 0.040) than in the baseline area (mean = 0.247; SD = 0.055), but the time series is highly variable throughout the period of image capture. There is a prominent seasonal peak in NDVI for both regions for each of the 10 years of observation (2000–2009) superimposed upon a longer term trend of NDVI decline over the period of record due to the effects of drought. From this uncalibrated time series with high variation, it is difficult to identify NDVI responses to flooding.

Fig. 2B shows the NDVI anomaly calculated by comparing wetland and baseline NDVI. In general, NDVI anomaly values are around zero, indicating that NDVI in the wetland and baseline areas are being influenced by the same variables i.e. there is no significant

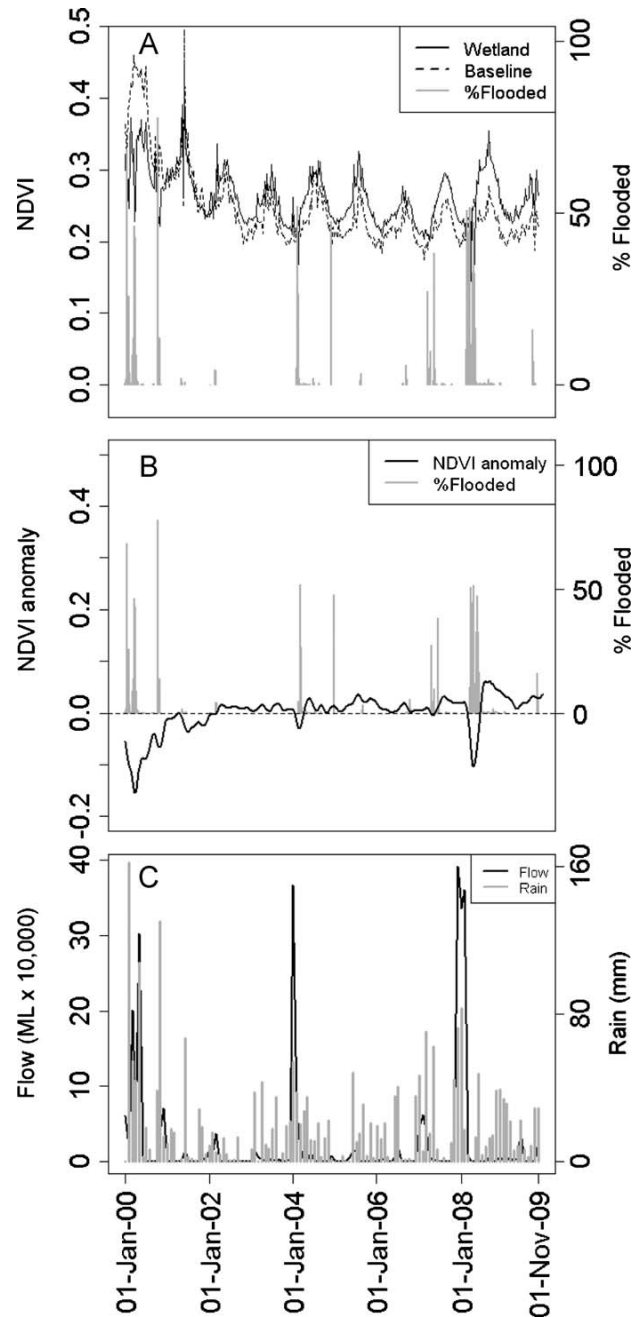


Fig. 2. NDVI and climate data for the Paroo River Wetlands showing (A) NDVI time series for the wetlands and the baseline area along with the proportion of the Paroo River Wetlands flooded, (B) Wetland vegetation response to flooding (NDVI anomaly) calculated by comparing the wetland and baseline NDVI, along with the proportion of the wetland flooded, and (C) total monthly channel flow in the Paroo River at Willara Crossing (Gauge 424002) and rainfall in the region of the Paroo River Wetlands (average of monthly total rainfall at Hungerford and Wanaaring, cf. Fig. 1).

effect due to flooding. The reduction in NDVI anomaly below zero between February 2000 and April 2002, and during flooding in June 2004 and October 2008, is a spectral artefact arising from the strong absorption of near infra-red (NIR) wavelengths by open water (Kirk, 1994). This decline is part of the normal trend of NDVI during inundation and does not indicate a change in ecological functioning.

The negative NDVI anomaly in the early part of the record before April 2002 coincides with flooding in the wetlands and the highest total monthly rainfall in this region during the period of image capture (Fig. 2C). This is the only time when NDVI is higher in the baseline region than on the wetlands (Fig. 2A). Plant productivity in the baseline region may have been responding to rainfall which, combined with the decline in NDVI of the floodplain due to inundation, gives the baseline region a relatively higher NDVI than the wetlands during this period. Periods of positive NDVI anomaly occur following flood recession (Fig. 2B) in the first half of 2004 and 2007 and, in particular, following a prolonged period of inundation of more than 50% of the wetlands in 2008. This latter flood coincided with total monthly flows exceeding 30,000 ML for several months, in addition to heavy rainfall (Fig. 2C), which resulted in the longest duration of inundation over a large extent of the floodplain in the period of record.

Several metrics can be calculated from the NDVI anomaly curve. NDVI anomaly values following the 2008 flood were positive between December 2007 and January 2009 (Fig. 2B) indicating that plant productivity in the wetland was increased above normal levels for approximately 13 months after this flood. Fig. 3 shows wetland NDVI for this period, together with the proportion of the wetland flooded. The upper dashed line shows the maximum NDVI during this period (0.355) and the lower dashed line shows maximum NDVI minus the maximum NDVI anomaly for this event (0.067). This indicates the maximum NDVI that might have occurred at this time in the absence of flooding. One interpretation of this is a peak increase in NDVI following flooding of approximately 19% over that which would have occurred from climatic and seasonal factors alone.

4. Discussion

The method presented in this paper highlights vegetation response to flooding in noisy time series of NDVI observations. Following the collection of appropriate field data it may be possible to calibrate NDVI anomaly metrics into units of biomass or leaf area index (LAI), as has been done for NDVI in other environments (Fuller, 1998; Gamon et al., 1995; Hill et al., 2004). This would enable the effectiveness of environmental flows to be reported in

terms of kilograms of biomass or change in foliar coverage per ML of water released, for example. While these data should not be used as the sole basis for environmental risk assessments or decisions on the management of environmental water, we think they would be a valuable supplement to monitoring programmes designed to assess the effectiveness of environmental flows in regulated river systems.

Calibrating biomass or LAI models from these data is not a trivial task, however, because the relationship between plant productivity and NDVI is complex. In general, higher NDVI values are correlated with higher productivity, but the representativeness of NDVI as a metric of vegetation biomass and growth vigour is subject to constraints. NDVI saturates at high biomass levels, above which it is no longer sensitive to increases in biomass or plant growth and vigour (Mutanga and Skidmore, 2004; Asner et al., 2005). NDVI is also sensitive to soil background characteristics including colour (Barry et al., 2008), age (Asner et al., 2005) and wetness (Rao et al., 1993). These factors can result in different NDVI values for two adjacent pixels in which vegetation biomass and growth vigour are effectively identical. In addition, the averaging effect of pixels, in which each pixel value represents an aggregation of all of the features visible within the area of the pixel, means that an identical NDVI value may be produced by a small amount of highly vigorous vegetation or a large amount of vegetation of lower vigour. Consequently, data describing parameters such as LAI and foliage projective cover (FPC), along with soil colour, texture and moisture measurements may be required. A number of other indexes complementary to the NDVI are also available, including the Enhanced Vegetation Index which may increase sensitivity to canopy structural changes in areas of higher biomass (Gao et al., 2000). Despite its constraints, NDVI remains the most commonly used remotely sensed metric of vegetation 'condition' because of its ease of calculation, its correlation with many parameters of vegetation structure and function and because its limitations are well characterised.

The results presented in this paper are necessarily influenced by the character of the datasets used. The processing of MODIS image datasets is constantly being reviewed and revised collections of existing images, which improve the accuracy of the data, are produced on occasion (<http://modis-land.gsfc.nasa.gov/news.htm>). Pixels in the MOD09 data used for NDVI and inundation detection are 500 m × 500 m in size and represent the average of all the land surface features detected within the pixel area. This can affect the interpretation of the condition of land surface features that are smaller than a pixel or highly spatially detailed. Similarly, the OWL algorithm for water detection is refined occasionally, and the 500 m pixel size means that pixels are identified as water only if it is sufficiently deep to be detected over a certain proportion of the pixel. Small or patchy inundation patterns may be missed from this analysis.

An important caveat when applying the method described in this manuscript is that, while the high temporal variation of plant productivity of floodplain and wetland ecosystems can make it difficult to observe the specific influence of an individual flood, this variability is a vital characteristic of wetland ecosystem function. Periods of low productivity and dryness are as important to the persistence of these systems as are large floods (van Oorschot et al., 1998; Colloff and Baldwin, 2010). Drying of the floodplain enables aerobic processes to occur within sediments which can facilitate new plant growth and increase nutrient availability upon rewetting (van Oorschot et al., 1998). Conditions under which floodplain ecosystem functions tend to be sub-optimal include persistent inundation, or severe or prolonged drought, with the highest productivity occurring in areas which are flooded moderately frequently (Sims and Thoms, 2002). Consequently, the importance of considering each flood event within the flooding regime cannot be overstated. Sub-optimal functioning might be indicated by a

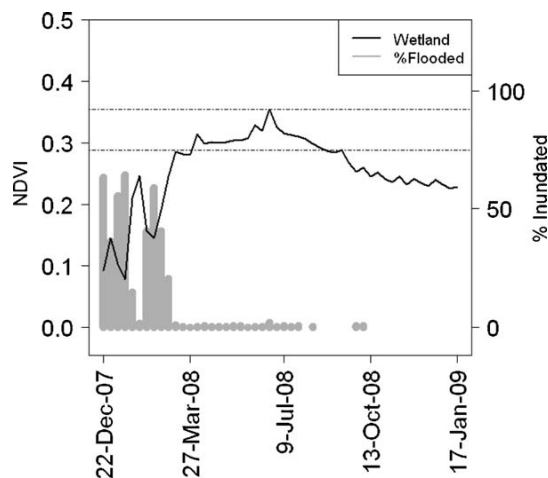


Fig. 3. Paroo River Wetlands NDVI between December 2007 and January 2009, showing the proportion of the wetlands inundated. The upper dashed line is the maximum NDVI observed during this flood response event (0.355). The lower dashed line shows maximum NDVI minus the maximum NDVI anomaly (0.067), and indicates the maximum NDVI that might have occurred at this time in the absence of flooding.

persistent decline in NDVI over the longer term or by substantially increased or reduced variability of NDVI over time (Rapport et al., 1985).

The primary limitation of this method is the requirement that a suitable baseline area, containing similar proportions of woodland and non-woodland cover, occurs close to the wetland of interest and within the same climatic zone. This was the case for Paroo River Wetlands but it was not possible to identify suitable baseline areas for several other important floodplain wetland ecosystems in the MDB. Methods to overcome this limitation are currently being investigated.

Acknowledgements

This research was funded by the National Water Commission as part of the project, 'Ecological Outcomes of Environmental Flows' and by CSIRO Water for a Healthy Country National Research Flagship. Many thanks to Glenn Newnham and Juan Pablo Guerschman for their comments on an earlier draft of this paper, and to Ian Overton, Darius Culvenor and Sadanandan Nambiar for support and assistance with completing this manuscript.

References

- Asner, G.P., Carlson, K.M., Martin, R.E., 2005. Substrate age and precipitation effects on Hawaiian forest canopies from spaceborne imaging spectroscopy. *Remote Sens. Environ.* 98, 457–467.
- Barry, K.M., Stone, C., Mohammed, C.L., 2008. Crown-scale evaluation of spectral indices for defoliated and discoloured eucalypts. *Int. J. Remote Sens.* 29, 47–69.
- Bodie, J.R., Semlitsch, R.D., 2000. Spatial and temporal use of floodplain habitats by lentic and lotic species of aquatic turtles. *Oecologia* 122, 138–146.
- Briggs, S.V., Thornton, S.A., 1999. Management of water regimes in River Red Gum *Eucalyptus camaldulensis* wetlands for waterbird breeding. *Aust. Zool.* 31, 187–197.
- Bunn, S.E., Davies, P.M., Winning, M., 2003. Sources of organic carbon supporting the food web of an arid zone floodplain river. *Freshw. Biol.* 48, 619–635.
- BRS, 2006. Land Use of Australia, Version 3 – 2001/2002. Bureau of Rural Sciences, Commonwealth of Australia, <http://data.australia.gov.au/92> (accessed 06.11.09).
- Chesterfield, E.A., 1986. Changes in the vegetation of the river red gum forest at Barmah, Victoria. *Aust. For.* 19, 4–15.
- Colloff, M.J., Baldwin, D.S., 2010. Resilience of floodplain ecosystems in a semi-arid environment. *Rangel. J.* 32, 305–314.
- Colloff, M., Jin, W., 2009. Floodplain classification. In: Overton, I.C., Colloff, M.J., Doody, T.M., Henderson, B., Cuddy, S.M. (Eds.), *Ecological Outcomes of Flow Regimes in the Murray-Darling Basin: Report prepared for the National Water Commission by CSIRO Water for a Healthy Country Flagship*. CSIRO, Canberra, p. 422.
- CSIRO, 2007. Water availability in the Paroo. In: *A Report to the Australian Government from the CSIRO Murray-Darling Basin Sustainable Yields Project*. CSIRO, Canberra.
- DCC, 2008. Forest Extent and Change (v4) – 1972–2006. Australian Government Department of Climate Change, National Carbon Accounting System, www.climatechange.gov.au/ncas/reports/tech09.html.
- DEWHA, 2005. Directory of Important Wetlands in Australia (DIWA) Spatial Database. Australian Government Department of the Environment, Water, Heritage and the Arts, http://asdd.ga.gov.au/asdd/tech/zap/advanced-full.zap?&target=ea-1&syntax=html&field1=any&field2=any&field3=any&field4=&field5=&field6=anzlic_search_word&field7=overlaps.bbox&term1=wetland&term2=water&term3=&term4=&term5=&term6=&term7=-9.00,138.00%20-29 (accessed 02.08.10).
- Donohue, R.J., McVicar, T.R., Roderick, M.L., 2009. Climate-related trends in Australian vegetation cover as inferred from satellite observations, 1981–2006. *Glob. Change Biol.* 15, 1025–1039.
- Fuller, D.O., 1998. Trends in NDVI time series and their relation to rangeland and crop production in Senegal, 1987–1993. *Int. J. Remote Sens.* 19, 2013–2018.
- Gamon, J.A., Field, C.B., Goulden, M.L., Griffen, K.L., Hartley, A.E., Joel, G., Penuelas, J., Valentini, R., 1995. Relationships between NDVI, canopy structure and photosynthesis in three Californian vegetation types. *Ecol. Appl.* 5, 28–41.
- Gao, X., Huete, A.R., Ni, W., Miura, T., 2000. Optical-biophysical relationships of vegetation spectra without background contamination. *Remote Sens. Environ.* 74, 609–620.
- Guerschman, J.-P., Sims, N.C., Warren, G., Arthur, A.D., Colloff, M.J., 2009. Remote sensing of inundation. In: Overton, I.C., Colloff, M.J., Doody, T.M., Henderson, B., Cuddy, S.M. (Eds.), *Ecological Outcomes of Flow Regimes in the Murray-Darling Basin: Report prepared for the National Water Commission by CSIRO Water for a Healthy Country Flagship*. CSIRO, Canberra, pp. 271–286.
- Hansen, P.M., Schjoerring, J.K., 2003. Reflectance measurement of canopy biomass and nitrogen status in wheat crops using normalized difference vegetation indices and partial least squares regression. *Remote Sens. Environ.* 86, 542–553.
- Hill, M.J., Donald, G.E., Hyder, M.W., Smith, R.C.G., 2004. Estimation of pasture growth rate in the south west of Western Australia from AVHRR NDVI and climate data. *Remote Sens. Environ.* 93, 528–545.
- IUCN, 2011. eFlowNet Summary 2010. International Union for Conservation of Nature, <http://cmsdata.iucn.org/downloads/newsletter-print.pdf> (accessed 14.06.11).
- Junk, W.J., Bayley, P.B., Sparks, R.E., 1989. The flood pulse concept in river-floodplain systems. *Can. Spec. Publ. Fish. Aquat. Sci.* 106, 110–127.
- Junk, W.J., Piedade, M.T.F., 1993. Biomass and primary-production of herbaceous plant communities in the Amazon floodplain. *Hydrobiologia* 263, 155–162.
- Kingsford, R.T., Lee, E., 2010. *Ecological Character Description of the Paroo River Wetlands Ramsar Site*. New South Wales Government Department of Environment Climate Change and Water, Sydney.
- Kirby, M., Van-Dijk, A.I.J.M., Mainuddin, M., Peña-Arancibia, J., Guerschman, J.P., Liu, Y., Marvanek, S., McJannet, D.L., Paydar, Z., McVicar, T.R., Van-Niel, T.G., Li, L.T., 2008. River water balance accounts across the Murray-Darling Basin 1990–2006. In: *A Report to the Australian Government from the CSIRO Murray-Darling Basin Sustainable Yields Project*. CSIRO, Canberra.
- Kirk, J.T., 1994. *Light and Photosynthesis in Aquatic Ecosystems*, second ed. Cambridge University Press, Cambridge.
- Mutanga, O., Skidmore, A.K., 2004. Narrow band vegetation indices overcome the saturation problem in biomass estimation. *Int. J. Remote Sens.* 25, 3999–4014.
- Ogden, R.W., Thoms, M., Levings, P.L., 2002. Nutrient limitation of plant growth on the Narran River, Australia: growth experiments and a pilot soil survey. *Hydrobiologia* 489, 277–285.
- Oindo, B.O., Skidmore, A.K., 2002. Interannual variability of NDVI and species richness in Kenya. *Int. J. Remote Sens.* 23, 285–298.
- Queensland Parks and Wildlife Service, 1999. Currawinya National Park Management Plan. Queensland Parks and Wildlife Service, Brisbane.
- Rao, P.V.N., Venkataratnam, L., Rao, P.V.K., Ramana, K.V., Singarao, M.N., 1993. Relation between root zone soil moisture and normalized difference vegetation index of vegetated fields. *Int. J. Remote Sens.* 14, 441–449.
- Rapport, D.J., Regier, H.A., Hutchinson, T.C., 1985. Ecosystem behaviour under stress. *Am. Nat.* 125, 617–640.
- Sheldon, F., Boulton, A.J., Puckridge, J., 2002. Conservation value of variable connectivity: aquatic invertebrate assemblages of channel and floodplain habitats of a central Australian arid-zone river, Cooper Creek. *Biol. Conserv.* 103, 13–31.
- Sims, N.C., Thoms, M.C., 2002. What happens when floodplains wet themselves: vegetation response to inundation on the Lower Balonne Floodplain. In: Dyer, F., Thoms, M.C., Olley, J.M. (Eds.), *Proceedings of the IAHS Symposium on the Structure, Function and Management Implications of Fluvial Sedimentary Systems*, Alice Springs. International Association of Hydrological Sciences, Wallingford, pp. 195–202.
- Sims, N.C., Chariton, A.A., Jin, H., Colloff, M.J., 2011. A regional classification of floodplains and wetlands of the Murray-Darling Basin based on changes in flows following river regulation and water resource development. *Wetlands*, doi:10.1007/s13157-011-0231-2.
- Thorp, J.H., Delong, M.D., 1994. The riverine productivity model: an heuristic view of carbon sources and organic processing in large river ecosystems. *Oikos* 70, 305–308.
- USGS, 2010. MOD09A1 Product Page. United States Geological Survey & NASA, Land Process Distributed Active Archive Centre MODIS Product Table, https://lpdaac.usgs.gov/lpdaac/products/modis_products_table/surface_reflectance/8_day_13_global_500m/mod09a1 (accessed 02.08.10).
- van Oorschot, M., Hayes, C., van Strien, I., 1998. The influence of soil desiccation on plant production, nutrient uptake and plant nutrient availability in two French floodplain grasslands. *Regul. Rivers. Res. Manag.* 14, 313–327.